

Creating NAT Gateways in AWS

Lab Details

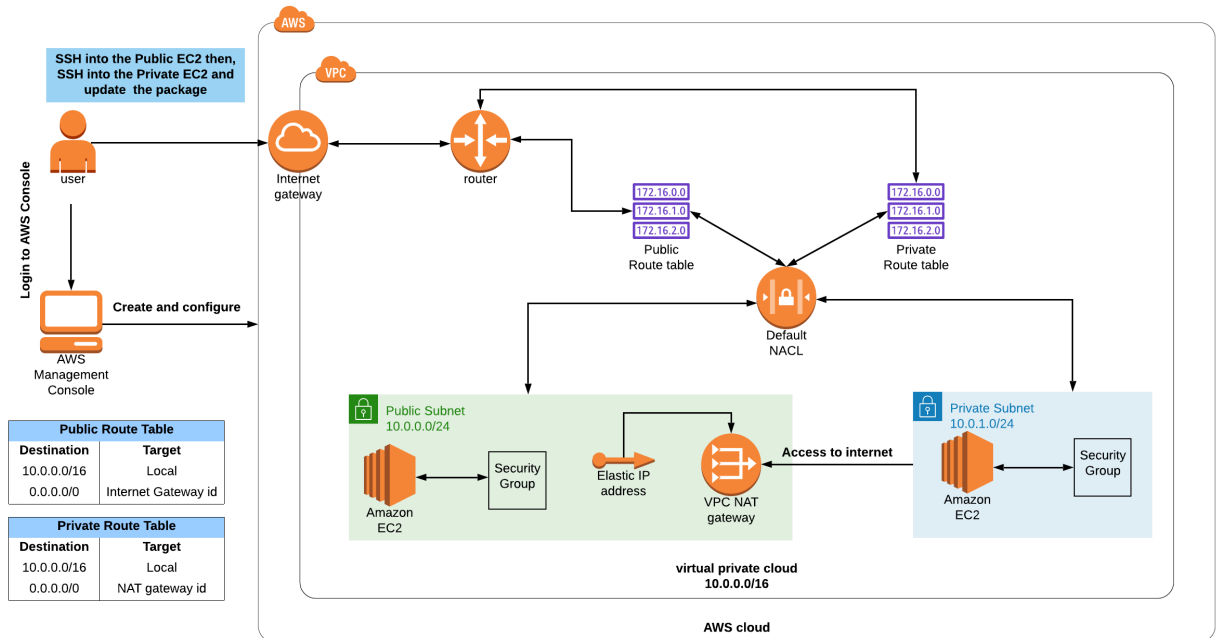
1. This lab walks you through the steps to create a NAT Gateway and allow internet access to an Instance in a private subnet.
2. Duration: **1 hour 30 minutes**
3. AWS Region: **US East (N. Virginia) us-east-1**

Introduction

What are NAT Gateways?

- NAT stands for Network Address Translation.
- A NAT Gateway is a device used to enable instances in a private subnet to connect to the internet or other AWS services.
- It prevents the internet from initiating connections with the instances present in the private subnet.
- It forwards traffic from the instance in the private subnet to the internet or other AWS services, and then sends the response back to the instances.
- Changes the instance's IP address with the NAT device's address when the traffic goes to the Internet.
- We have 2 kinds of NAT devices:
 - NAT Instance
 - NAT Gateway
- NAT Instance uses Amazon Linux AMIs.
- NAT Instance limit depends on your instance type limit for the region.
- NAT Instance does not support IPv6 traffic.
- NAT Gateway usage is charged to the customer on an hourly basis.
- NAT Gateway does not support IPv6 traffic.
- AWS recommends the usage of NAT Gateway, since they provide better availability and bandwidth over NAT Instances.

Architecture Diagram



Task Details

1. Log into AWS Management Console.
2. Create a VPC.
3. Create public and private subnets.
4. Create Internet Gateway
5. Create Public route table and configure
6. Launch an EC2 instance in Public subnet.
7. Launch an EC2 instance in Private subnet.
8. SSH into Public and Private EC2 instance and Test Internet Connectivity
9. Create a NAT Gateway
10. Update Route table and configure NAT Gateway
11. Test Internet connection from Instance inside Private Subnet
12. Validation of the lab.

Lab Steps

Task 1: Create VPC

1. Make sure you are in the **US East (N. Virginia) us-east-1** region.
2. Navigate to **VPC** under the services menu. Click on **Your VPCs**.
3. Click on **Create VPC**.
 - Name Tag : Enter **MyVPC**

- IPv4 CIDR block : Enter **10.0.0.0/16**(You can also put any other CIDR range)
 - IPv6 CIDR block : Select **No IPv6 CIDR Block**
 - Tenancy : **Default**
 - Click on **Create VPC**.
4. The VPC is now created.

Name	VPC ID	State	IPv4 CIDR
MyVPC	vpc-0aff084cd082b37ec	✔ Available	10.0.0.0/16

Task 2: Create Public and Private Subnets

1. Navigate to **Subnets** in the left panel of the VPC page.
2. Let's create a Public subnet. Click on **Create subnet**.
 - VPC ID : select **MyVPC**
 - Subnet Name : Enter **MyPublicSubnet**
 - Availability Zone : **No Preference**
 - IPv4 CIDR block : Enter **10.0.0.0/24**
 - Click on **Create subnet**.
3. Let's enable Auto Assign public IP to Instances created within this subnet,
 - Select **MyPublicSubnet** , Click on **Actions**.
 - Click on **Modify auto-assign IP settings**
 - Enable auto-assign public IPv4 address : **Check**
 - Click on **Save**.
4. Now the Instances launched inside the **MyPublicSubnet** will have Public IPs assigned to them by default.
5. Let's create a private subnet. Click on **Create subnet**.
 - VPC ID : select **MyVPC**
 - Subnet Name : Enter **MyPrivateSubnet**
 - Availability Zone : **No Preference**
 - IPv4 CIDR block : Enter **10.0.1.0/24**
 - Click on **Create subnet**.
6. Now, two subnets are created.

Subnets (8) Info				
Filter subnets				
☐	Name ▾	Subnet ID ▾	State ▾	VPC ▾
☐	-	subnet-5d047f01	✓ Available	vpc-3418934e Default VPC
☐	MyPublicSubnet	subnet-025aec644b5a387e4	✓ Available	vpc-0bc6ac5e4082bc2ba My...
☐	-	subnet-6d090727	✓ Available	vpc-3418934e Default VPC
☐	-	subnet-afb73591	✓ Available	vpc-3418934e Default VPC
☐	MyPrivateSubnet	subnet-06eb435482cdf40f4	✓ Available	vpc-0bc6ac5e4082bc2ba My...
☐	-	subnet-0141710e	✓ Available	vpc-3418934e Default VPC
☐	-	subnet-e54702cb	✓ Available	vpc-3418934e Default VPC
☐	-	subnet-d52759b2	✓ Available	vpc-3418934e Default VPC

Task 3: Create Internet Gateway

1. Navigate **Internet Gateways** in the left panel of the VPC page.
2. Click on **Create internet gateway**.
 - Name tag : Enter **MyIGW**
 - Click on **Create internet gateway**.
3. An Internet Gateway is now created.

☐	Name ▾	Internet gateway ID ▾	State ▾
☐	MyIGW	igw-04d05d03331d86a1f	⊖ Detached

4. To attach Internet Gateway to a VPC,
 - Select the Internet Gateway **MyIGW**.
 - Click on **Actions**. Select **Attach to VPC**.
 - VPC : Select **MyVPC**
 - Click on **Attach internet gateway**.
5. Now **MyIGW** is attached to **MyVPC**.

☐	Name ▾	Internet gateway ID ▾	State ▾	VPC ID
☐	MyIGW	igw-04d05d03331d86a1f	✓ Attached	vpc-070606d8d0c3b038a MyVPC

Task 4: Create Public Route Table and Configure

We will create a route table and attach a public subnet to it. Instances launched within this subnet will have access to the Internet.

1. Navigate to Route Table in the left panel of the **VPC** page.

2. Click on **Create route table**.
 - Name tag : Enter **PublicRouteTable**
 - VPC : Select **MyVPC**
 - Click on **Create route table**.
3. A route table by name **PublicRouteTable** will be created.
4. To attach an Internet Gateway, select **PublicRouteTable**.
5. In the **Routes** tab below:
 - Click on **Edit routes**.
 - On the next page, click on **Add route**.
 - Destination : Enter **0.0.0.0/0**
 - Target : Select **Internet Gateway**, and once the internet gateways have been created, select **MyIGW**
 - Click on **Save changes**.
6. To associate the Public Subnet to the route table, Select **PublicRouteTable**.
 - Click on the **Subnet Associations** tab.
 - Click on **Edit subnet associations**.
 - On the next page, select **MyPublicSubnet** from the list displayed.
 - Click on **Save associations**.
7. Once all the configurations are completed, it should look like below:

	Name	Route table ID	Explicit subnet associat...	Edge associations
☒	PublicRouteTable	rtb-0e3142c2be4db29c1	subnet-05e8088df36ba...	–
☐	–	rtb-3d998f42	–	–
☐	–	rtb-075ca0659c4417c2a	–	–

Filter routes			Both
Destination	Target	Status	
10.0.0.0/16	local	Active	
0.0.0.0/0	igw-04d05d03331d86a1f	Active	

8. Now the Instances launched within **MyPublicSubnet** will have access to the Internet.

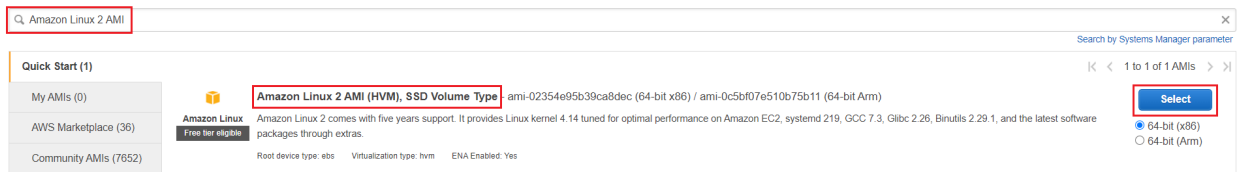
- As you can see, there is another existing route table already available for **MyVPC**. It is a main route table created at the time the VPC was created. We will use it while creating the **NAT Gateway**.

Task 5: Launch an EC2 Instance in Public Subnet

- Make sure you are in the **N.Virginia** region.
- Navigate to the **Services** menu in the top, click on **EC2** in the  **Compute** section.
- Navigate to **Instances** on the left panel and click on **Launch instances**.
- Choose an Amazon Machine Image (AMI): Search for **Amazon Linux 2 AMI** in the search box and click on the **select** button.

Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace, or you can select one of your own AMIs.



- Choose an Instance Type: select **t2.micro** **Free tier eligible** and click on the **Next: Configure Instance Details**
- Configure Instance Details:
 - Network : Enter **MyVPC**
 - Subnet : Enter **MyPublicSubnet**
 - Auto-assign Public IP : Select **Use Subnet Setting(Enable)** - default
 - Leave all other settings as default.

- Click on **Next: Add Storage**

- Add Storage: No need to change anything in this step. Click on

Next: Add Tags

- Add Tags: Click on **Add Tag**
 - Key : Enter **Name**
 - Value : Enter **MyPublicEC2Server**

- Click on **Next: Configure Security Group**

9. Configure Security Group:

- Assign a security group : Select **Create a new security group**
- Security group name : Enter **MyEC2Server_SG**
- Description : Enter **Security Group to allow traffic to EC2**
- To add **SSH**
 - Choose Type: Select **SSH**
 - Source: Select **Anywhere**
- Click on **Review and Launch**

10. Review and Launch : Review all settings and click on

Launch

11. Key Pair : Create a new key pair and click on

Download Key Pair

- Name KeyPair as **PublicEC2key**
- Key Pair Type as **RSA**

Launch Instances

12. Click on

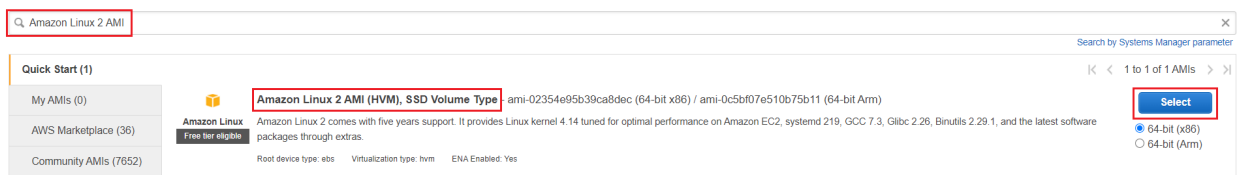
13. Launch Status: Your instance is now launching, Select the instance and wait for it to change status to **Running**.

Task 6: Launch an EC2 Instance in Private Subnet

1. Click on **Launch instances**.
2. Choose an Amazon Machine Image (AMI): Search for **Amazon Linux 2 AMI** in the search box and click on the **select** button.

Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace, or you can select one of your own AMIs.



3. Choose an Instance Type: select **t2.micro** and click on the

Next: Configure Instance Details

4. Configure Instance Details:

- Network : Select **MyVPC**
- Subnet : Select **MyPrivateSubnet**
- Auto-assign Public IP : Select **Use Subnet Setting(Disable)** - default
- Leave all other settings as default.

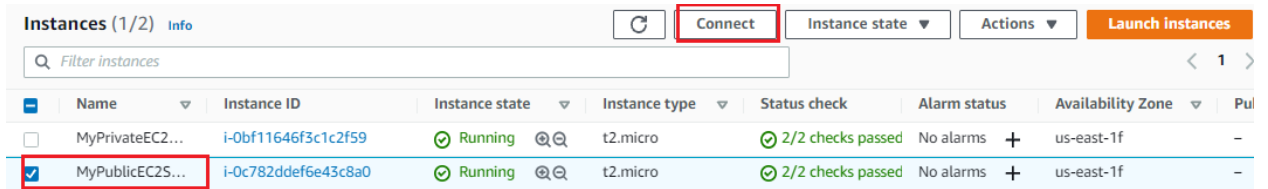
- Click on **Next: Add Storage**.
5. Add Storage: No need to change anything in this step. Click on **Next: Add Tags**.
6. Add Tags: Click on **Add Tag**.
- Key : Enter **Name**
 - Value : Enter **MyPrivateEC2Server**
 - Click on **Next: Configure Security Group**.
7. Configure Security Group:
- Assign a security group: Choose **Select an existing security group**
 - Select **MyEC2Server_SG** from the list
 - Click on **Review and Launch**.
8. Review and Launch : Review all settings and click on **Launch**.
9. Key Pair : Create a new key pair and click on **Download Key Pair**.
- Name KeyPair as **MyKey**
 - Key Pair Type as **RSA**
10. Click on **Launch Instances**.
11. Launch Status: Your instance is now launching, Select the instance and wait for it to change status to **Running**.
12. Note the Private IP Address of **MyPrivateEC2Server** : Example 10.0.1.45
13. Two servers are now launched and ready.

☐	Name ▼	Instance ID	Instance state ▼	Instance type ▼
☐	MyPublicEC2S...	i-030ad1e8acba40553	✓ Running	t2.micro
☐	MyPrivateEC2...	i-08c80170ce8923fab	✓ Running	t2.micro

Task 7: SSH into Public and Private EC2 Instance and Test Internet Connectivity

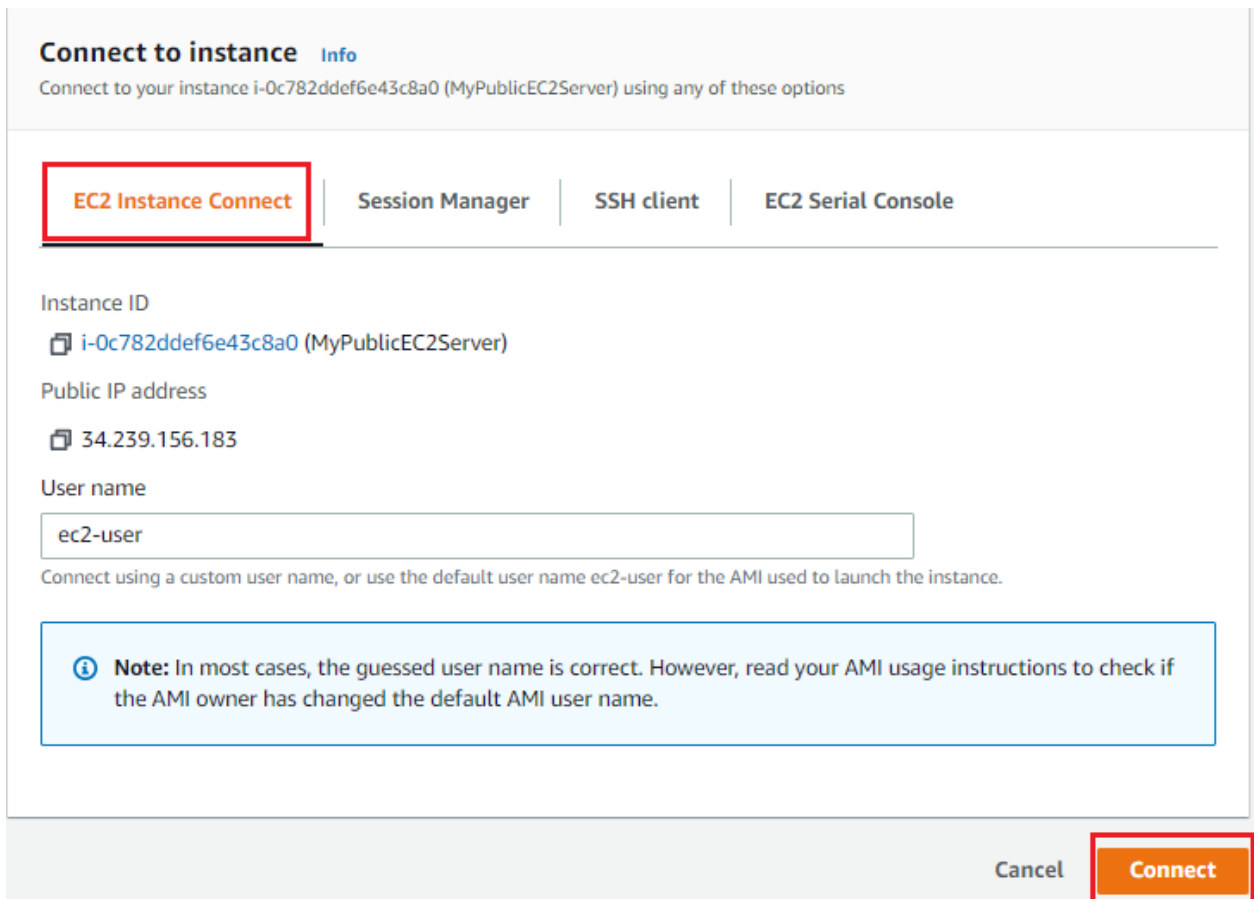
1. SSH into **MyPublicEC2Server** Instance. Follow the below steps

- Once the instance is created. Select the instance **MyPublicEC2Server**. Click on connect button



Instances (1/2) Info									
Filter instances									
	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone		
☐	MyPrivateEC2...	i-0bf11646f3c1c2f59	Running	t2.micro	2/2 checks passed	No alarms	us-east-1f	+	-
☒	MyPublicEC2S...	i-0c782ddef6e43c8a0	Running	t2.micro	2/2 checks passed	No alarms	us-east-1f	+	-

- Now select the EC2 instance connect option. You can see there are four options for connecting to EC2. You can use any of the given options to get in the console but for our lab, we are using EC2 instance connect.



Connect to instance Info

Connect to your instance i-0c782ddef6e43c8a0 (MyPublicEC2Server) using any of these options

EC2 Instance Connect

Session Manager

SSH client

EC2 Serial Console

Instance ID
i-0c782ddef6e43c8a0 (MyPublicEC2Server)

Public IP address
34.239.156.183

User name

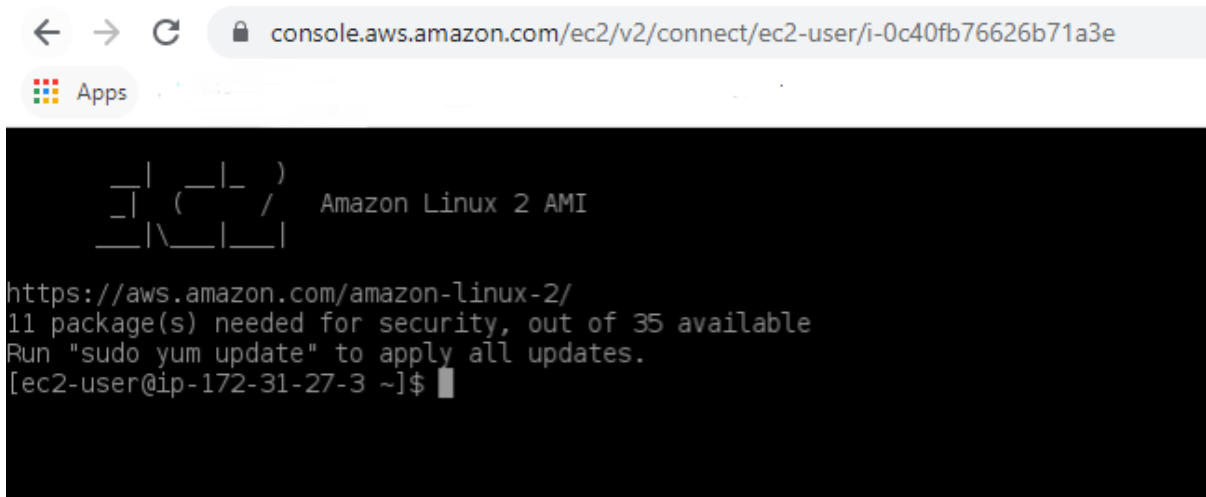
Connect using a custom user name, or use the default user name ec2-user for the AMI used to launch the instance.

Note: In most cases, the guessed user name is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI user name.

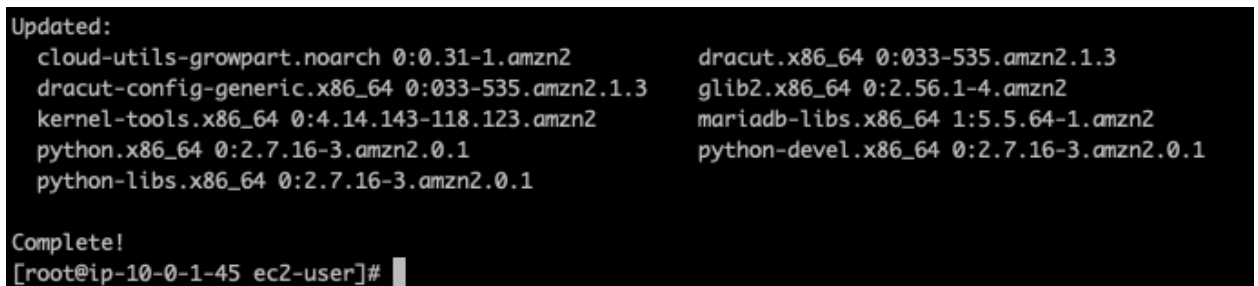
Cancel

Connect

- A new tab would be opened in your browser where we can see the console.



- Switch to root user: `sudo su`
- Run the updates using the following command:
 - `yum -y update`
- Since the Internet Gateway **MyIGW** is connected to **MyPublicSubnet**, updates will be completed successfully.



- Let's SSH into **MyPrivateEC2Server** from **MyPublicEC2Server**.
- In order to SSH into **MyPrivateEC2Server**, first, we need to create the PEM file in the public EC2 ie, **MyPublicEC2Server**, and copy the data from our **MyKey.pem** in the local machine.
- We need **MyKey.pem** in order to SSH. We shall copy key details from the **MyKey.pem** from your local machine (which was downloaded earlier while launching EC2 instances).
- To copy the contents of the **MyKey.pem**, open the file in a text editor and copy the whole content.
- To create the **MyKey.pem** in **MyPublicEC2Server**, run
 - `vi MyKey.pem`
- Now press the following for inserting the data(You can see insert popped at the last then you can paste the key).
 - Click `i`

```
-- INSERT --
```

Note: In the editor, copy and paste the key that looks similar to the example below:

```
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAmSHG3lriawEHo93izlRP4Ll0wE8SdGlxabrnIe3BUyRy5A05
ciELGww06TJihcogH30lMZvMXKxA+9fEXYMnwVGMYhOzELisZqmh5A7nD12H+Osc
wcMmahaj4tQsL3fK0aZp0jKMwsLm3XTDkmMnR/IrSUGGS5b8uY0d/91Ew550+e5p
dgWShnYxvG+LPHdIRrJqymbmGqwzQoZlsqTv+rMqMVLZB8J3JZo0gRib0cqH13iX
n9BwoYANw4xDivBL0RjDTC6freArAMZ2hvyLYrnSY0qFM+8Kic/AgZ0GNDLt2B0m
6R34oQmmR2GsGRawabLGwyIFIdILAXaC40aLDwIDAQABAoIBAQCkxw6hQ3sZ6MtD
Ja1IPc9LxKD/DBm0jyghGnY+tMjS2DK3Sqe4Mu2UmaGdeXNkareXvMr8y6vdxxKX
RQxwLzbFY22yZmDus8zw3Qz4TFPVB6n0Npw37vJEvhwZrgg8Rn0G591waKfmXwS
R61Bzuzg1cf/b2UQ0Btd+Sts0w3ixNys8tLPbwgZTYXSoq6MstCd0utQ+ggSduGR
I7wyLIQAiwAI5d7TRKr3pGV6mdTxQdRfiXnLh0xNzH6qDYt8TqFMipAGTMnCK+ET
zY2EgXY3TX/9CE1o85wi0ho2BtL7eDp4ZcksEmws1YzIAaXL01HZZEn7N+J850YS
XlZ9G1HhAoGBAPd6+11ay4keXx0w9hS9wGnVkaIBS8bLtf0J0Ig2RA5YFL30FxVw
MIrHslY5QC1SHofST7RGeSj0rc5cWpw4/byByro/jvEn1S2y8sBFz7SPa+jPK2dE
CKn9IGkFDXn15fQ9/u2gCVf10dbh5sFD6mwYo1JeX5oGcRHo8ScbcnypAoGBAKAV
ohIrNBRxB74LHHI5rjoHu7KLYn4MZ689I5y7AwFmVi57KmBqj7EowacNhaaLk0Yd
F5Aem9Itj60c/OMwai9cN2zpDuFqHTNfIoXit4dtCUhshz+qCxEqy/MmlkB52KmN
2b2+A4rtu0V95db0++k0R4r8wzwe+93uzVy01KT3AoGBANHdoLwR+vhUoSQIGSaN
SyS0z8MqhjAXIEVsXUI9gokHeM9crYaSiKenoTwtYJc0RIw5e43NFYHdgtaoysIg
MkxhZKF8tKMcvFL8pY8XpIPh/Ah7G0uBbjd5SLrGZh8xLkhpKn4ocMPxSHmJKD5M
cccpPL3M+hWLO0b300mGpRfBAoGATcx1qXLJqq2L37YyquiRXa5ob8sBoNSShz2L
UwQnFZjQhWJiI91l2+k8w9z0eQ4w6150FxxpCeo0i0ktNZRP5phgb10MYxaSJ8R
fhPCsdQx0Lq9uBkBg+Qqng7AsrEEtNI1Yds4iYoic5wN37vLuXgo1Mjz4s1TCnaF
42oWdd0CgYBOy+PpIIZNJjw6nIYvqD0oKn1AjICzbXv4YlHwYuVrH92TVUurPISe
7XN9nUJdztBemkzEL5SZtSeXuISmdqPjb+/2vg07xjIhYGEwwMDMuKLRfAlgA5/I
ENX7o9RuYJ+EySq9w23XveQuFMohJnA1RGeEbAt76BTidYZQCg0sfQ==
-----END RSA PRIVATE KEY-----
```

-- INSERT --

14. Save the File

- click **esc**
- **:wq**

```

fhPCsdQx0Lq9uBkBq+Qqng7AsrEEtNI1Yds4iYoic5wN37vLuXgo1Mjz4s1TCnaF
42owdd0CgYB0y+PpIIZNJjW6nIYvqDQoKn1AjICzbXv4YlHwYuVrH92TVUurpISe
7XN9nUJdztBemkzEL5SZtSeXuISmdqPjb+/2vg07xjIhYGEwwMDMuKLRFA1gA5/I
ENX7o9RuYJ+EySq9w23XveQuFMohJnA1RGeEbAt76BTidYZQCg0sfQ==
-----END RSA PRIVATE KEY-----
~
~
~
~
~
:wq

```

14. Check that the file was created correctly.

- `ls`

```

[root@ip-10-0-0-18 ec2-user]# ls
MyKey.pem
[root@ip-10-0-0-18 ec2-user]#

```

16. Update Permissions for the MyKey.pem

- `chmod 400 MyKey.pem`

17. Use the Private IP address of **MyPrivateEC2Server** to SSH.

- `ssh ec2-user@10.0.1.93 -i MyKey.pem`
- **Note:** In case if this message shows **Are you sure you want to continue connecting (yes/no)?** : Enter **yes**

```

[root@ip-10-0-0-18 ec2-user]# chmod 400 MyKey.pem
[root@ip-10-0-0-18 ec2-user]# ssh -i "MyKey.pem" ec2-user@10.0.1.45

  _ | _ | _ )
  _ | (   /   Amazon Linux 2 AMI
  _ | \  | _ |

https://aws.amazon.com/amazon-linux-2/
[ec2-user@ip-10-0-1-45 ~]$

```

18. Switch to root user

- `sudo su`

19. Run the updates using the following command:

- `yum -y update`

```

[root@ip-10-0-1-45 ec2-user]# yum -y update
Failed to set locale, defaulting to C
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd

```

20. Since no internet access is provided for EC2 instances in a private subnet, you will not be able to get updates. After some time, it will fail with the following message.

```
slower). If it is a very temporary problem though, this is often a nice
compromise:

yum-config-manager --save --setopt=<repoid>.skip_if_unavailable=true

Cannot find a valid baseurl for repo: amzn2-core/2/x86_64
[root@ip-10-0-1-45 ec2-user]#
```

Task 8: Create a NAT Gateway

NAT Gateway is used to provide internet access to the Instances inside a private subnet.

1. Navigate to the VPC Page.
2. Make sure you are still in the **N.Virginia** Region.
3. In the Left Panel, click on **NAT Gateways**.
4. Click on **Create NAT gateway**.
 - Name : Enter **MyNATGateway**
 - Subnet : Choose **MyPublicSubnet** in **MyVPC**
 - Elastic IP Allocation ID: Click on **Allocate Elastic IP**.
 - Once the new Elastic IP is allocated, click on **Create NAT gateway**.
5. Note that NAT Gateway is always created in a public subnet.
6. NAT Gateway will be created in a few minutes. Once created, the status will change to **available**.

	Name ▾	NAT gateway ID ▾	State ▾	State message ▾
🔵	MyNATGateway	nat-0dd9d8d0fd9481414	🟢 Available	-

Task 9: Update Route table and configure NAT Gateway

1. Navigate to **Route Tables** in the left panel.
2. You can see two Route Tables available for **MyVPC**

☐	Name ▾	Route table ID ▾	Explicit subnet associat... ▾	Edge associations ▾	Main ▾	VPC ▾
☐	PublicRouteTable	rtb-0e3142c2be4db29c1	subnet-05e8088df36ba...	-	No	vpc-070606d8d0c3b038a MyVPC
☐	-	rtb-3d998f42	-	-	Yes	vpc-15048f6f Default VPC
☐	-	rtb-075ca0659c4417c2a	-	-	Yes	vpc-070606d8d0c3b038a MyVPC

3. To attach Nat Gateway, select the Main Route Table (which is different from the one created by you).

4. In the **Routes** tab below,

1. Click on **Edit routes**.
2. In the next page, Click on **Add route**.
3. Destination: Enter **0.0.0.0/0**
4. Target: Select **NAT Gateway**, and once the internet gateways have loaded, select the NAT Gateway you created.
5. Click on **Save changes**.

5. Once all the configurations are completed, it should look like below.

	Name	Route table ID	Explicit subnet associat...	Edge associations	Main	VPC
☐	PublicRouteTable	rtb-0e3142c2be4db29c1	subnet-05e8088df36ba...	–	No	vpc-070606d8d0c
☐	–	rtb-3d998f42	–	–	Yes	vpc-15048f6f De
☒	–	rtb-075ca0659c4417c2a	–	–	Yes	vpc-070606d8d0c

Routes (2)			
Filter routes		Both	
Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	nat-0dd9d8d0fd9481414	Active	No

6. Now the Instances launched within **MyPrivateSubnet** can access the Internet through the NAT Gateway.

Task 10: Test Internet connection from Instance inside Private Subnet

1. SSH back into **MyPublicEC2Server**.
2. Switch to root user
 - `sudo su`
3. SSH into **MyPrivateEC2Server**
 - `ssh -i "MyKey.pem" ec2-user@10.0.1.45`
4. Switch to root user
 - `sudo su`
5. Run the updates using the following command:
 - `yum -y update`
6. You can see that the updates have been completed successfully in the terminal.

```
Updated:
cloud-utils-growpart.noarch 0:0.31-1.amzn2
dracut-config-generic.x86_64 0:033-535.amzn2.1.3
kernel-tools.x86_64 0:4.14.143-118.123.amzn2
python.x86_64 0:2.7.16-3.amzn2.0.1
python-libs.x86_64 0:2.7.16-3.amzn2.0.1
dracut.x86_64 0:033-535.amzn2.1.3
glib2.x86_64 0:2.56.1-4.amzn2
mariadb-libs.x86_64 1:5.5.64-1.amzn2
python-devel.x86_64 0:2.7.16-3.amzn2.0.1

Complete!
[root@ip-10-0-0-18 ec2-user]#
```

7. This shows that **MyPrivateEC2Server** has internet access.
8. Use `exit` command to close the private server connection.

Completion and Conclusion

- You have successfully created a new VPC from scratch and created both public and private subnets.
- You have created an Internet Gateway and configured a new route table.
- You have launched 1 EC2 instance each of the Public and Private subnets and tested Internet access from them.
- To provide Internet access to the EC2 instance in the Private subnet, you created a NAT Gateway and configured a Route table.
- You confirmed that the instance in the private subnet is able to connect to the internet.